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EXPERIMENT 1

EXPERIMENT 1.1

Aim: Write a problem statement for your IPD project.

Title: Plant Disease Detection

Problem Statement:

Diseases in plants cause a threat to the yield and are a significant proportion why the global yield has reduced. Many a times, farmers themselves are not well educated in this field and hence, even after knowing that the crop has been affected with some disease, there is not much they can do to save the crop. Traditional methods are not very accurate and may also take a lot of time.

Plant disease detection application is a Machine Learning based application which detects common crop diseases. Users have to click a photo of the affected part of the plant and upload it to the app. Based on the uploaded image, the app will give the possible diagnosis based on the symptoms visible in the image and even provide treatment options. The application will provide a detailed report on the detected disease and available treatment options in an understandable format even to people who do not have knowledge about plant diseases. There will also be an e-commerce section in the app where users can shop for fertilizers, medicines and other products required for maintaining plant health. The app will have multiple language options to support diverse users.

Plagiarism Report:

English
French
Spanish
German

Problem Statement:

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198 Words

Analysis complete

EXPERIMENT 1.2

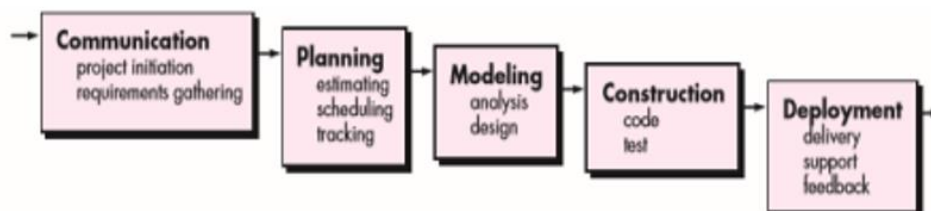
Aim: Describe various process models with diagrams.

Process Models:

1) Waterfall Model

The Waterfall Model is one of the most traditional approaches to project management, particularly in software development. It is a linear and sequential process, meaning that each phase of the project must be completed before moving to the next one. The stages typically include requirement gathering, system design, implementation, testing, deployment, and maintenance. Once a phase is finished, there is no going back, making this model suitable for projects with clearly defined requirements that are unlikely to change. However, it's less flexible when dealing with evolving needs and can lead to challenges if issues arise late in the project lifecycle.

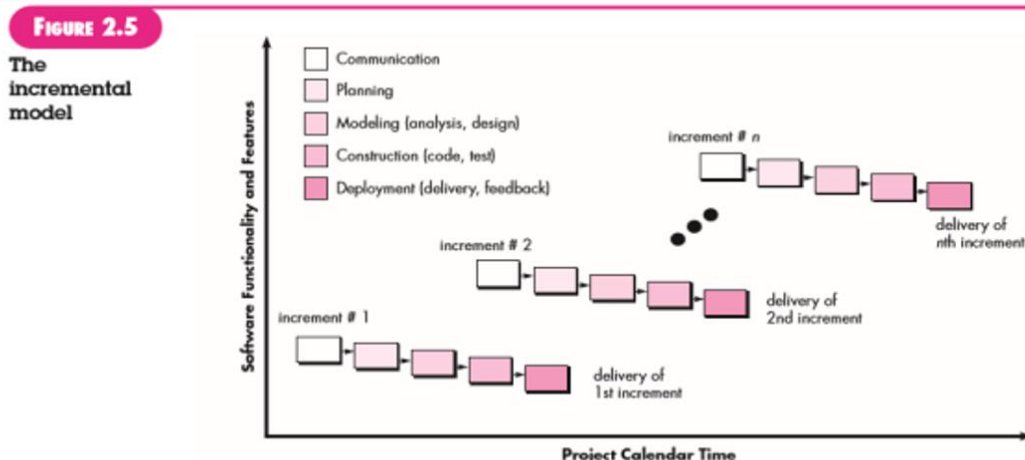
FIGURE 2.3 The waterfall model



- Advantages of waterfall model-
 - This model works for small projects because the requirements are understood very well.
 - The waterfall model is simple and easy to understand, implement, and use.
 - All the requirements are known at the beginning of the project, hence it is easy to manage.
- Disadvantages of the waterfall model
 - The problems with this model are uncovered, until the software testing.
 - The amount of risk is high.
 - This model is not good for complex and object oriented projects.

2) Incremental Process Model

The incremental model combines elements of linear and parallel process flows. Referring to Figure 2.5, the incremental model applies linear sequences in a staggered fashion as calendar time progresses. Each linear sequence produces deliverable “increments” of the software. When an incremental model is used, the first increment is often a core product. That is, basic requirements are addressed but many supplementary features (some known, others unknown) remain undelivered. The core product is used by the customer (or undergoes detailed evaluation). As a result of use and/or evaluation, a plan is developed for the next increment. The plan addresses the modification of the core product to better meet the needs of the customer and the delivery of additional features and functionality. This process is repeated following the delivery of each increment, until the complete product is produced. The incremental process model focuses on the delivery of an operational product with each increment. Early increments are stripped-down versions of the final product, but they do provide capability that serves the user and also provide a platform for evaluation by the user.



- Advantages of incremental model
 - This model is flexible because the cost of development is low and initial product delivery is faster.
 - It is easier to test and debug during the smaller iteration.
 - The working software generates quickly and early during the software life cycle.
 - The customers can respond to its functionalities after every increment.
- Disadvantages of the incremental model
 - The cost of the final product may cross the cost estimated initially.
 - This model requires a very clear and complete planning.
 - The planning of design is required before the whole system is broken into small increments.
 - The demands of customer for the additional functionalities after every increment causes problem during the system architecture.

3) Evolutionary Model

It is also called successive versions model or incremental model.

At first, a simple working model is built.

Subsequently it undergoes functional improvements & we keep on adding new functions till the desired system is built.

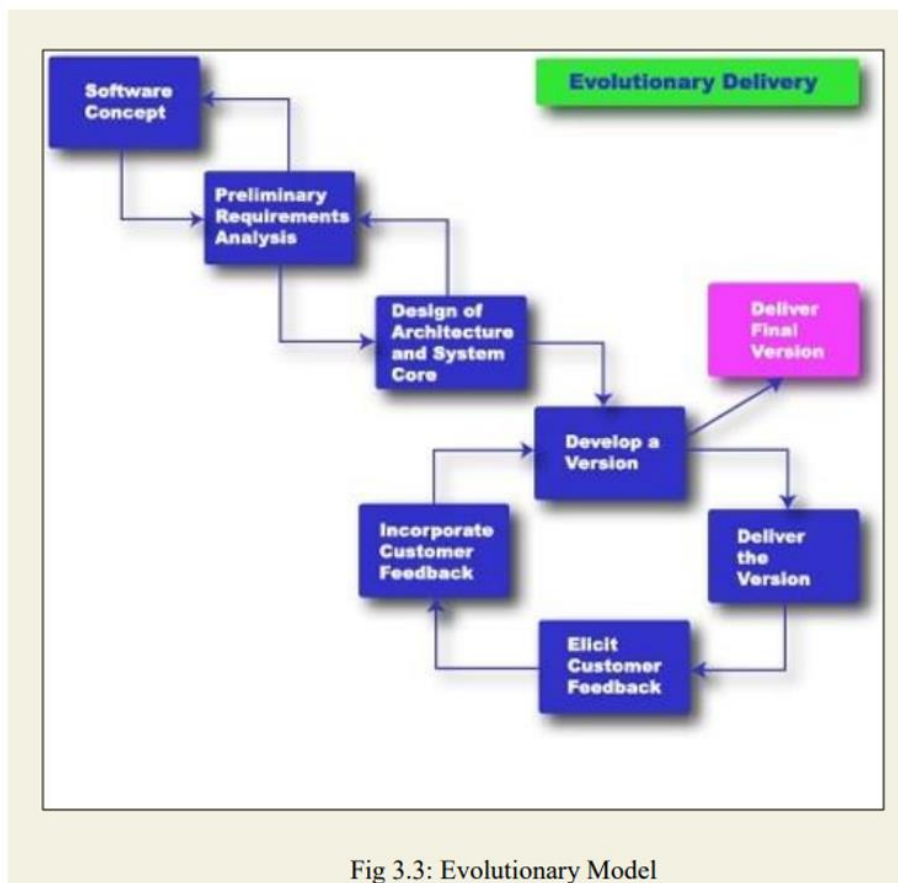


Fig 3.3: Evolutionary Model

- Advantages:
 - User gets a chance to experiment partially developed system.
 - Reduce the error because the core modules get tested thoroughly.

- Disadvantages:
 - It is difficult to divide the problem into several versions that would be acceptable to the customer which can be incrementally implemented & delivered.

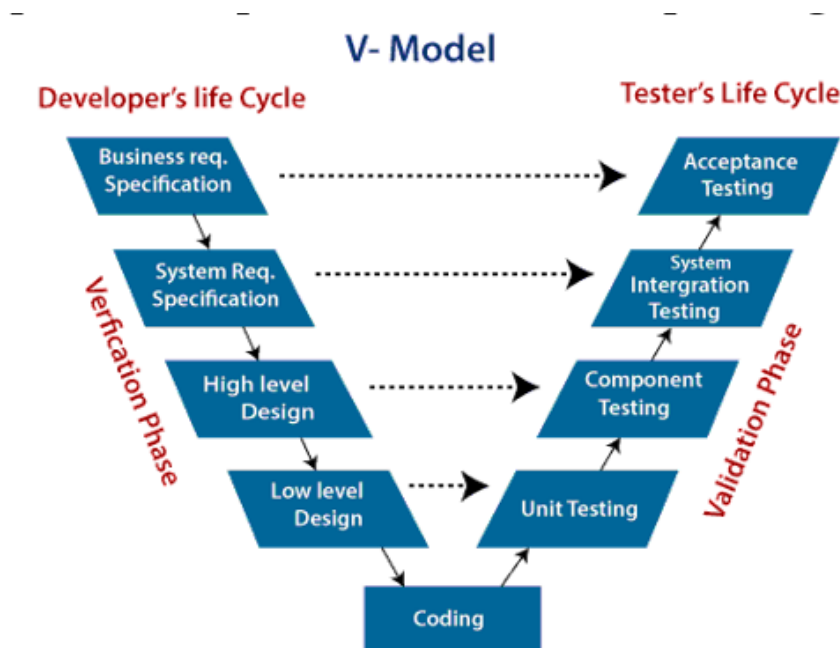
4) V Model

V-Model also referred to as the Verification and Validation Model.

In this, each phase of SDLC must complete before the next phase starts. It follows a sequential design process same as the waterfall model.

Testing of the device is planned in parallel with a corresponding stage of development.

- **Verification:** It involves a static analysis method (review) done without executing code. It is the process of evaluation of the product development process to find whether specified requirements meet.
- **Validation:** It involves dynamic analysis method (functional, non-functional), testing is done by executing code. Validation is the process to classify the software after the completion of the development process to determine whether the software meets the customer expectations and requirements.
- So V-Model contains Verification phases on one side of the Validation phases on the other side. Verification and Validation process is joined by coding phase in V-shape.
- Thus it is known as V-Model.



5) Concurrent Model

Concurrent engineering is an approach to product development in which multiple phases of the product development process are carried out simultaneously instead of sequentially. The main objective of concurrent engineering is to improve product quality and shorten development time by overlapping the different product life cycle stages. Concurrent engineering uses tools like development models to maximize the product development Process. Concurrent engineering is a holistic approach to product

development in which all stakeholders work together from the beginning to the end of the project.

Concurrent engineering is like a team sport because it requires close collaboration and communication among all members of the development team and the customer.

Each team member must know what the other members are working on and how their work fits into the project.

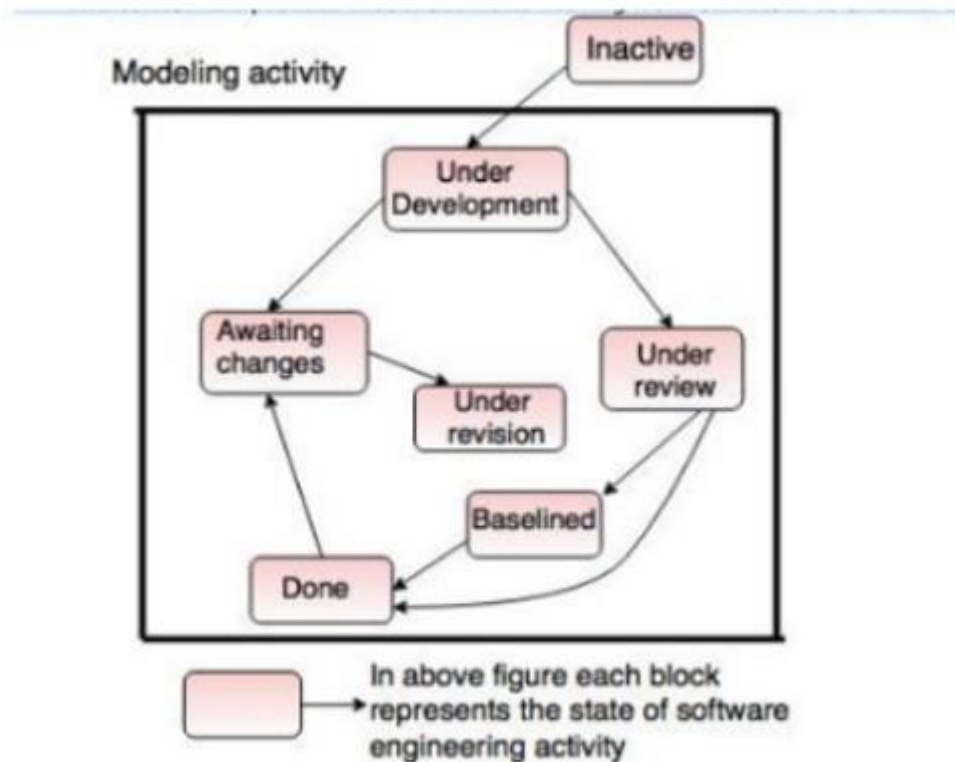


Fig. - One element of the concurrent process model

- Advantages of the concurrent development model
 - This model is applicable to all types of software development processes.
 - It is easy for understanding and use.
 - It gives immediate feedback from testing.
 - It provides an accurate picture of the current state of a project.
- Disadvantages of the concurrent development model
 - It needs better communication between the team members. This may not be achieved all the time.
 - It requires to remember the status of the different activities.

6) Spiral Model

Spiral model is a risk driven process model.

It is used for generating the software projects.

In spiral model, an alternate solution is provided if the risk is found in the risk analysis, then alternate solutions are suggested and implemented.

It is a combination of prototype and sequential model or waterfall model.

In one iteration all activities are done, for large project's the output is small.

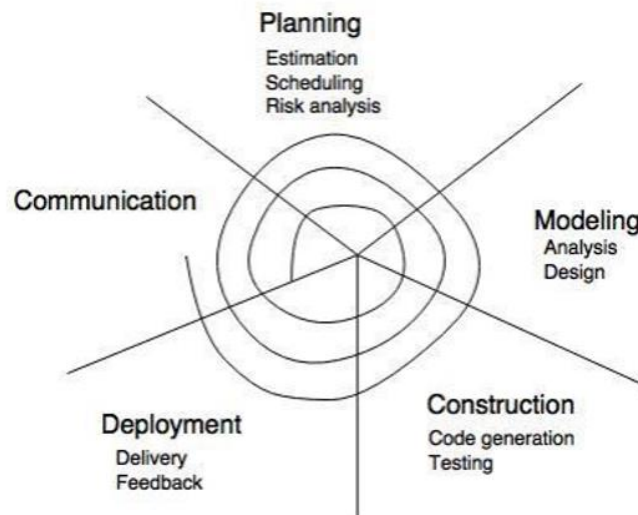


Fig. - The Spiral Model

- Advantages of Spiral Model
 - It reduces high amount of risk.
 - It is good for large and critical projects.
 - It gives strong approval and documentation control.
 - In spiral model, the software is produced early in the life cycle process.
- Disadvantages of Spiral Model
 - It can be costly to develop a software model.
 - It is not used for small projects.

7) Agile Model

The meaning of Agile is swift or versatile. "Agile process model" refers to a software development approach based on iterative development. Agile methods break tasks into smaller iterations, or parts do not directly involve long term planning. The project scope and requirements are laid down at the beginning of the development process. Plans regarding the number of iterations, the duration and the scope of each iteration are clearly defined in advance.

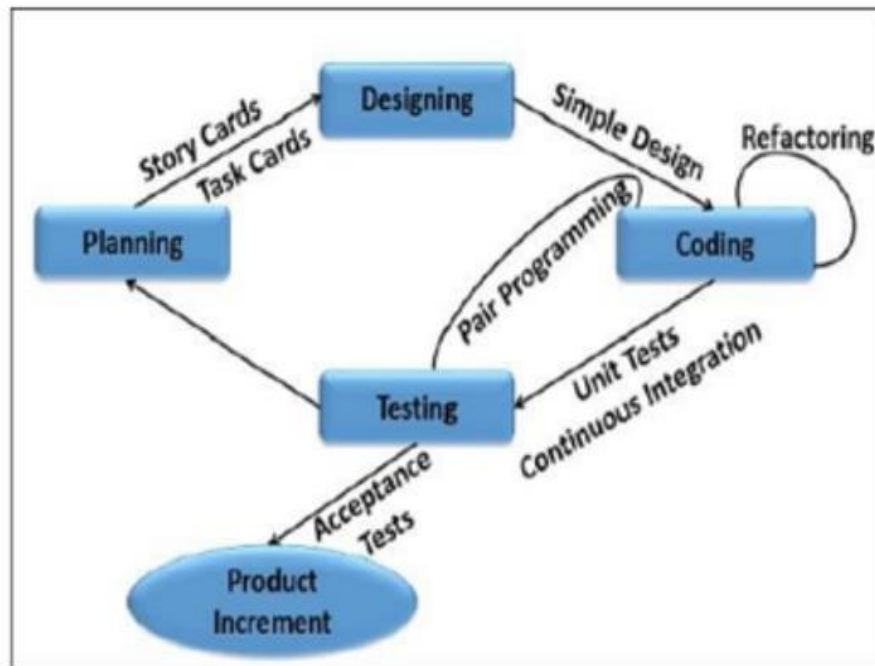


Fig. Agile Model

- Advantages:
 - Frequent Delivery
 - Face-to-Face Communication with clients.
 - Efficient design and fulfils the business requirement.
 - Anytime changes are acceptable.
 - It reduces total development time.
- Disadvantages:
 - Due to the shortage of formal documents, it creates confusion and crucial decisions taken throughout various phases can be misinterpreted at any time by different team members.
 - Due to the lack of proper documentation, once the project completes and the developers allotted to another project, maintenance of the finished project can become a difficulty.

8) Extreme Programming (XP)

XP is a lightweight, efficient, low-risk, flexible, predictable, scientific way to develop software. Extreme Programming (XP) was conceived and developed to address the specific needs of software development by small teams in the face of vague and changing requirements. Extreme Programming is one of the Agile software development methodologies. It provides values and principles to guide the team behavior. The team is expected to self-organize. Extreme Programming provides specific core practices where each practice is simple and self-complete. Combination of practices produces more complex and emergent behavior.



9) Scrum

Scrum is aimed at sustaining strong collaboration between people working on complex products, and details are being changed or added. It is based upon the systematic interactions between the three major roles: Scrum Master, Product Owner, and the Team. Scrum Master is a central figure within a project. His principal responsibility is to eliminate all the obstacles that might prevent the team from working efficiently. Product Owner, usually a customer or other stakeholder, is actively involved throughout the project, conveying the global vision of the product and providing timely feedback on the job done after every sprint. Scrum Team is a cross-functional and self-organizing group of people that is responsible for the product implementation. It should consist of up to 7 team members, in order to stay flexible and productive.

SCRUM PROCESS



7) Kanban Model

Kanban is a popular framework which is used to implement agile software development. It takes real time communication of capacity and complete transparency of work. The work items are represented in a kanban board visually, allowing team members to see the state of every piece of work at any time.

Requirement / Task / Incident Progress					
Backlog	Planned	In Progress	Developed	Tested	Completed
User Story	User Story TK TK TK	User Story	TK TK	User Story TK	User Story TK TK
User Story	IN	User Story TK	TK TK IN	TK	IN IN
User Story		IN			
User Story					
User Story					

Model we used in our IPD:

Agile Model

Our app is ML based. ML models need continuous improvements based on testing. Flexibility to add new features like disease prediction accuracy improvements or better dataset handling. Regular feedback loop (you can test with farmers/agriculture experts).

Conclusion:

Thus we successfully prepared detailed statement of problem for the selected IPD project and identify suitable process model for the same with justification.



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EXPERIMENT 2

Aim: Develop Software Requirement Specification (SRS) document in IEEE format for the project.

Software Requirements Specification (SRS) for Agriculture App

Version: 1.0

Date: 19.02.2025

Prepared by: Gargi S

1. Introduction

1.1 Purpose

This document defines the functional, non-functional, and system requirements for the agriculture app, which provides **a community-driven platform with chatbot assistance and multilingual support** to help farmers collaborate and make informed decisions.

1.2 Document Conventions

This document follows the IEEE 830-1998 standard for SRS documentation.

1.3 Intended Audience and Reading Suggestions

- **Developers:** For understanding system design and API integration.
- **Project Managers:** To track project scope and deliverables.
- **Stakeholders (Farmers, Agronomists, Agri-businesses):** To ensure user needs are met.

1.4 Product Scope

The app provides **community engagement, AI chatbot support, and multilingual accessibility** for better farming decisions.

2. Overall Description



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2.1 Product Perspective

- **Standalone Mobile/Web Application**
- **Integration with AI Chatbot & Cloud-based Data Processing**

2.2 Product Functions

1. **User Registration & Authentication**
2. **Community Forums & Groups**
3. **AI Chatbot Assistance**
4. **Multilingual Support**
5. **Weather API Integration**
6. **Crop & Soil Health Monitoring**

2.3 User Characteristics

- **Farmers:** Need simple UI & voice support.
- **Agronomists:** Require advanced analytics.
- **Admins:** Manage users & moderate discussions.

2.4 Constraints

- **Internet Dependency**
- **Device Compatibility**
- **Data Privacy Compliance**

3. Specific Requirements



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3.1 Functional Requirements

ID	Requirement	Description
FR-1	User Authentication	Users log in via email/phone, password, or biometric authentication.
FR-2	Community Forum	Users can create, join, and participate in discussions.
FR-3	AI Chatbot	Provides instant farming advice and troubleshooting.
FR-4	Multilingual Support	Users select their preferred language for UI & chatbot responses.
FR-5	Weather API Integration	Fetches and displays real-time weather data.
FR-6	Crop & Soil Health Monitoring	Users input crop details for AI-based recommendations.

3.2 Non-Functional Requirements

- **Performance:** Load time ≤ 3 seconds.
- **Security:** AES-256 encryption for user data.
- **Scalability:** Support [X] concurrent users.

3.3 External Interface Requirements

- **APIs:** Weather API, AI Chatbot API, Translation API.
- **UI/UX:** Must follow **Material Design (Android) / Human Interface Guidelines (iOS)**.

4. System Architecture & Diagrams

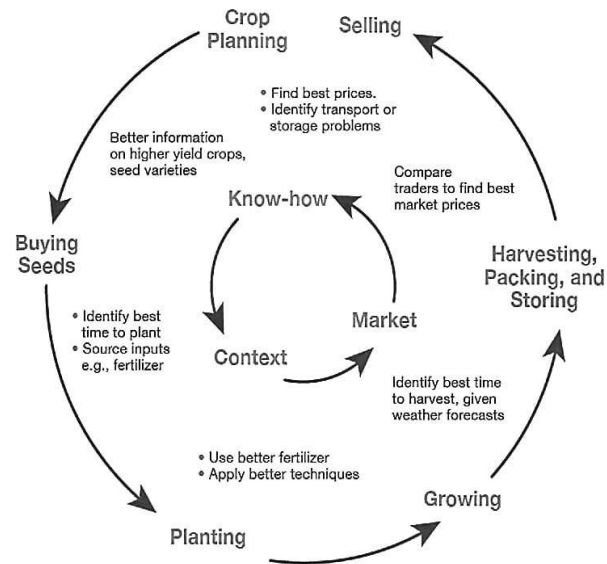
This section provides a visual representation of the system architecture, workflows, and data flow.

4.1 Proposed System

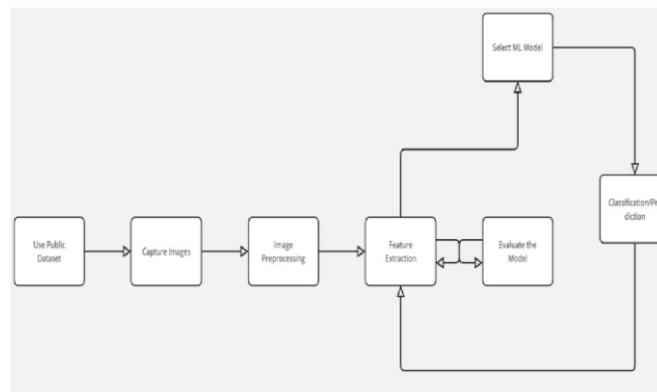


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4.5 Flowchart



5. Appendices

- **References:**

- **Standards & Guidelines:**

- IEEE 830-1998 – Recommended Practice for Software Requirements Specifications.
- ISO/IEC 25010:2011 – Software Quality Model.
- Agile Manifesto – <https://agilemanifesto.org> (For Agile Development).

- **APIs & External Services:**



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- OpenWeather API – <https://openweathermap.org/api> (For real-time weather updates).
- Google Translate API – <https://cloud.google.com/translate> (For multilingual support).
- AI Chatbot API (e.g., OpenAI, Dialogflow) – [Insert API documentation link].

➤ **Frameworks & Technologies Used:**

- React Native / Flutter – Mobile app development frameworks.
- Node.js / Django / Spring Boot – Backend technologies.
- Firebase / PostgreSQL / MongoDB – Database solutions.

• **Glossary:**

Term	Definition
AI Chatbot	An artificial intelligence-based system that provides automated responses and farming assistance.
API (Application Programming Interface)	A set of protocols and tools that allow different software applications to communicate.
Authentication	The process of verifying a user's identity before granting access to the system.
Backend	The server-side logic and database management that handle requests from the frontend.
Community Forum	A feature where users can post discussions, ask questions, and interact with other farmers and agronomists.
Crop & Soil Health Monitoring	The process of analyzing crop conditions and soil health using AI-based recommendations.
DFD (Data Flow Diagram)	A visual representation of how data moves through the system.
Frontend	The part of the application that users interact with, such as mobile or web UI.
Iteration	A cycle in Agile development where features are built, tested, and improved based on feedback.
Multilingual Support	The ability of the app to provide content and chatbot responses in multiple languages.
Sprint	A time-boxed development cycle in Agile methodology where a specific set of features is built and tested.
System Architecture	The design and structure of the software system, including components and their interactions.



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UAT (User Acceptance Testing)	The process of testing the application with actual users to ensure it meets their requirements.
Weather API	An external service that provides real-time weather data to the application.

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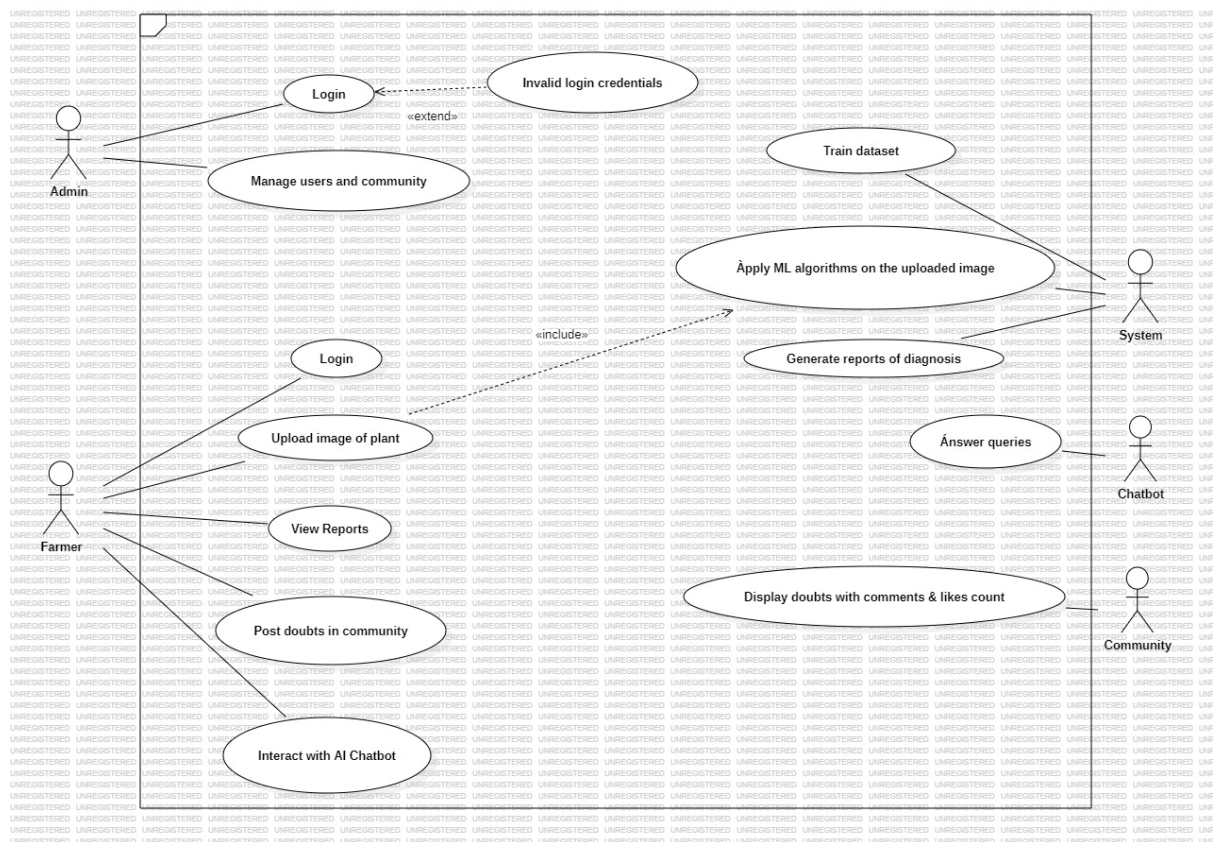
EXPERIMENT 3

Aim: Identify Actors, use cases and scenarios draw Use case diagram and Class diagram. draw use case and class diagram for IPD project.

Project Title: AgriSmart

AgriSmart is a smart agriculture app designed to help farmers with essential tools and insights. It features weather forecasting, soil health analysis, and AR-based pest & disease identification to optimize farming. The app also provides market price tracking, a farming calendar, and a wholesaler dashboard for better crop management. A community forum and AI-powered chatbot assist farmers with queries, while offline mode ensures accessibility in remote areas. With multilingual support (English, Hindi, Marathi) and yield prediction & analytics, AgriSmart empowers farmers with data-driven solutions for smarter and more efficient farming.

Use Case Diagram:



Class diagram:

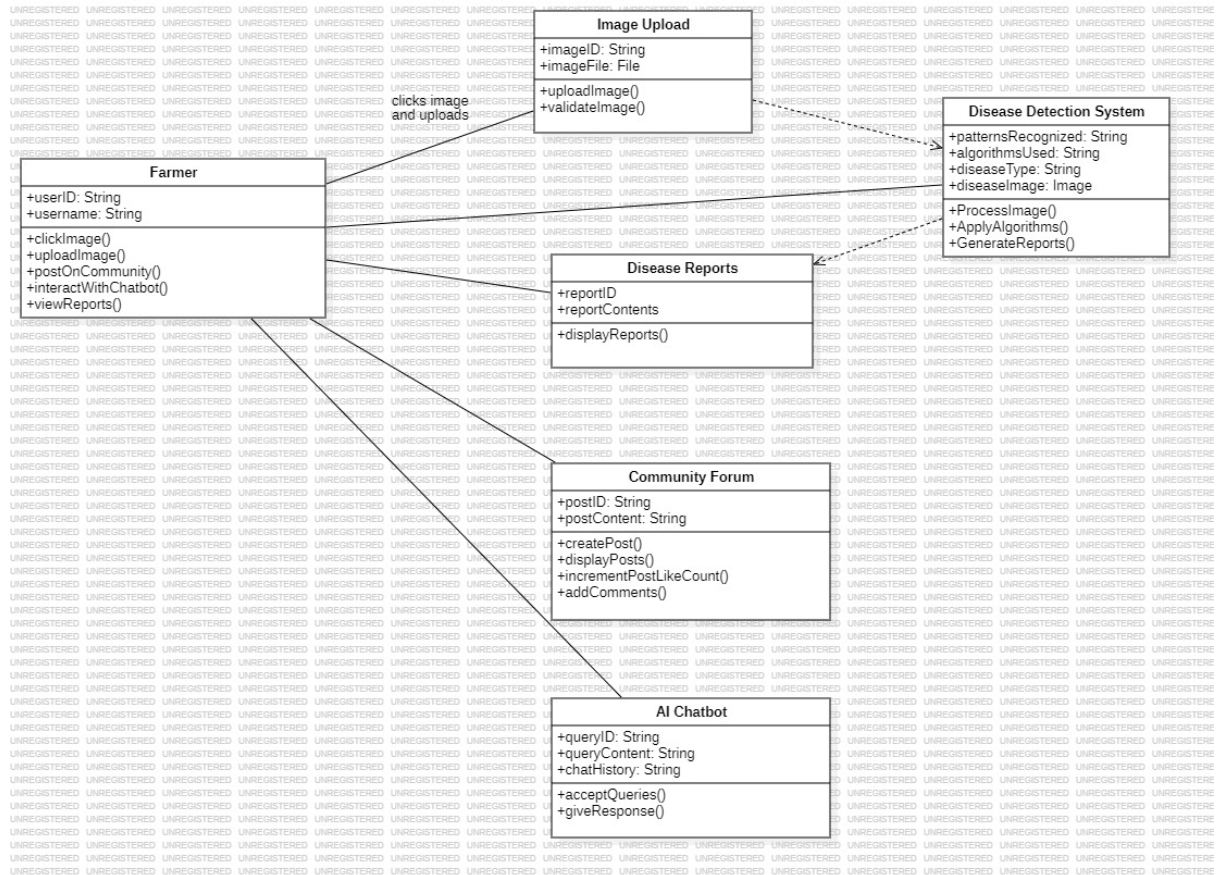


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Actors:

- Farmer (user)
- Admin
- System
- Chatbot
- Community

Use cases:

• Admin:

- Login (with an extension for invalid login credentials)
- Manage users and community

• Farmer:

- Login (with an extension for invalid login credentials)
- Upload an image of a plant
- View Reports



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- Post doubts in the community
- Interact with AI Chatbot

• **System:**

- Apply ML algorithms on the uploaded image (includes generating reports of diagnosis)
- Train dataset

• **Chatbot:**

- Answer queries

• **Community:**

- Display doubts with comments & likes count

Scenarios:

- **Admin logs in** to manage users and the community.
- **Farmer logs in** to access system features.
- **Farmer uploads an image of a plant**, and the system applies ML algorithms to generate a diagnosis report.
- **Farmer views the generated report** after diagnosis.
- **Farmer posts doubts in the community**, which are displayed with comments and like counts.
- **Farmer interacts with the AI chatbot**, which answers queries.
- **The system trains its dataset** to improve future diagnoses.

Conclusion:

Thus we successfully drew use case and class diagrams for our IPD project and identified the actors, use cases and scenarios.



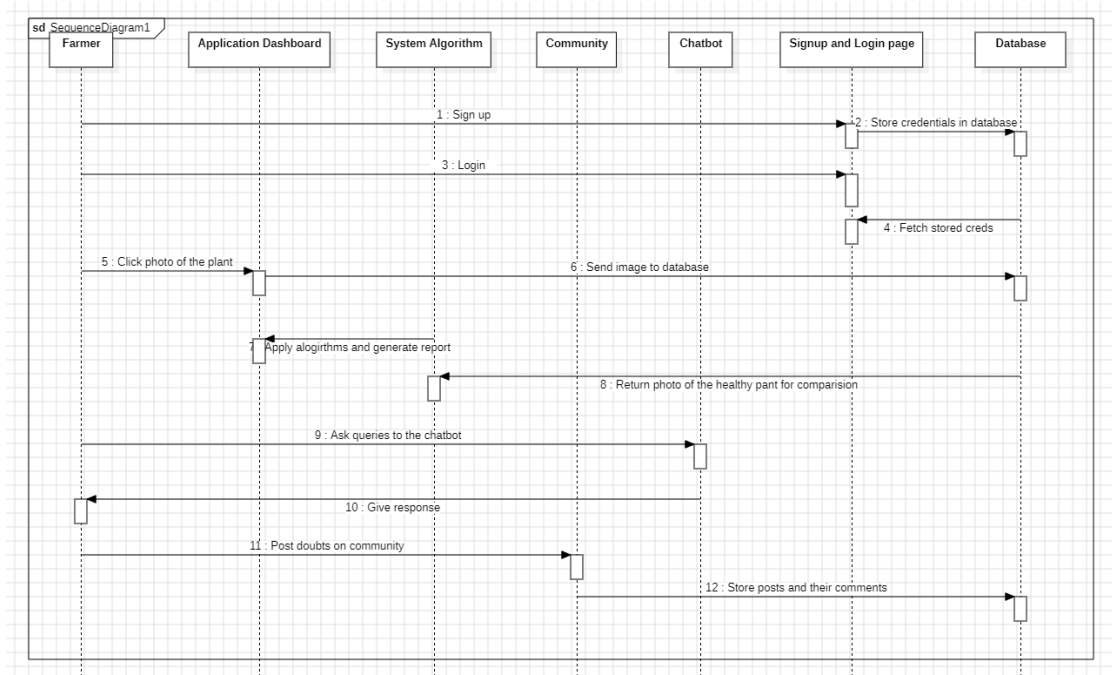
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EXPERIMENT 4

Aim: To draw Sequence and collaboration diagram.

1) Sequence diagram



Components:

- Actors:
 - Farmer
 - Application dashboard
 - System Algorithm
 - Community
 - Chatbot
 - Signup and Login page
 - Database
- Messages:
 - Signup
 - Store credentials in database
 - Login
 - Fetch stored credentials

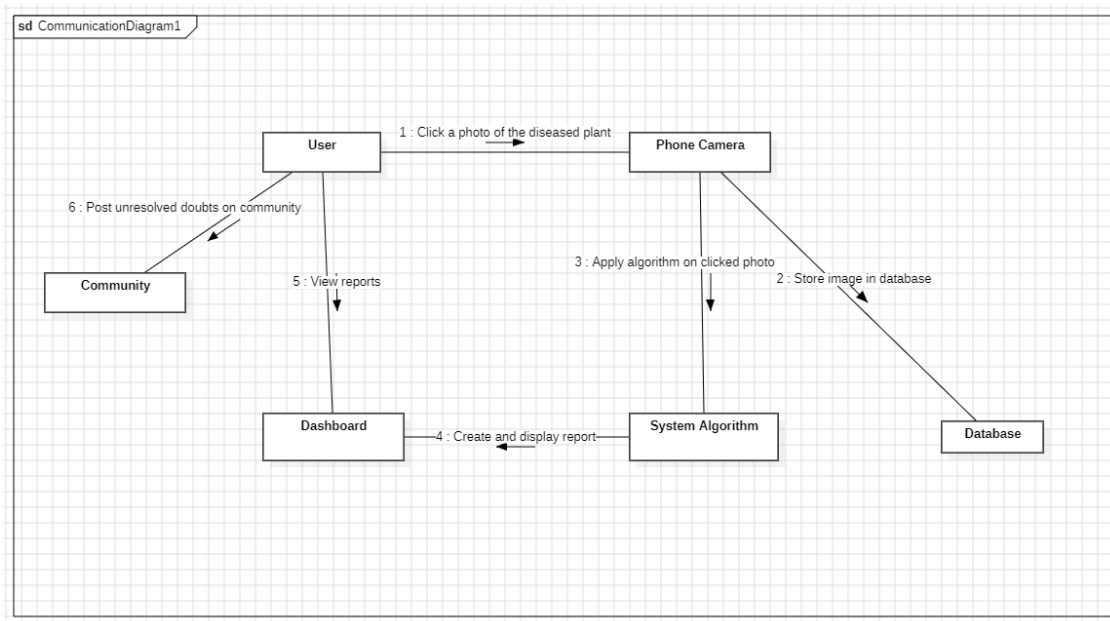


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- Click photo of the plant
- Send image to the database
- Apply algorithms and generate report
- Return photo of healthy plant for comparison
- Ask queries to chatbot
- Give response
- Post doubts on the community
- Store posts and their comments

2) Collaboration diagram



Components:

- Objects:
 - User
 - Phone camera
 - Dashboard
 - System Algorithm
 - Community
 - Database
- Messages:
 - Click a photo of the plant
 - Store image in database
 - Apply algorithms on clicked photo
 - Create and display reports



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- View reports
- Post unresolved doubts on community

Conclusion:

Thus we successfully drew Sequence and collaboration diagram for our IPD project.



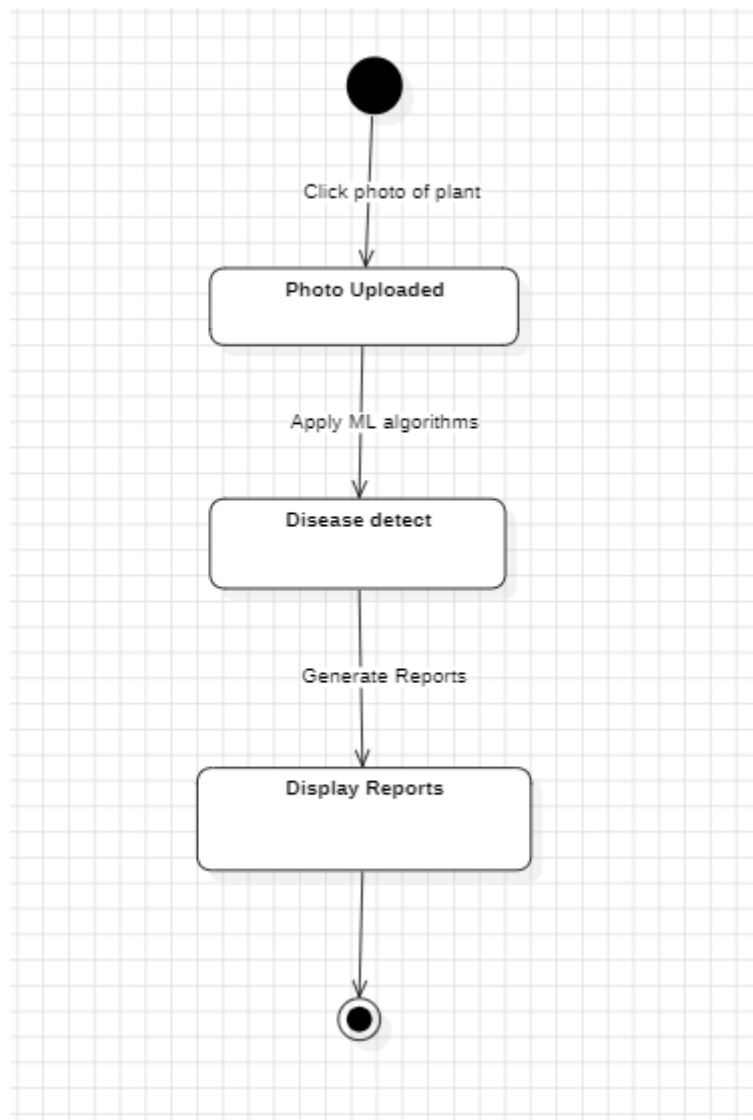
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EXPERIMENT 5

Aim: Develop state transition and activity diagram.

State transition Diagram:



Activity Diagram:

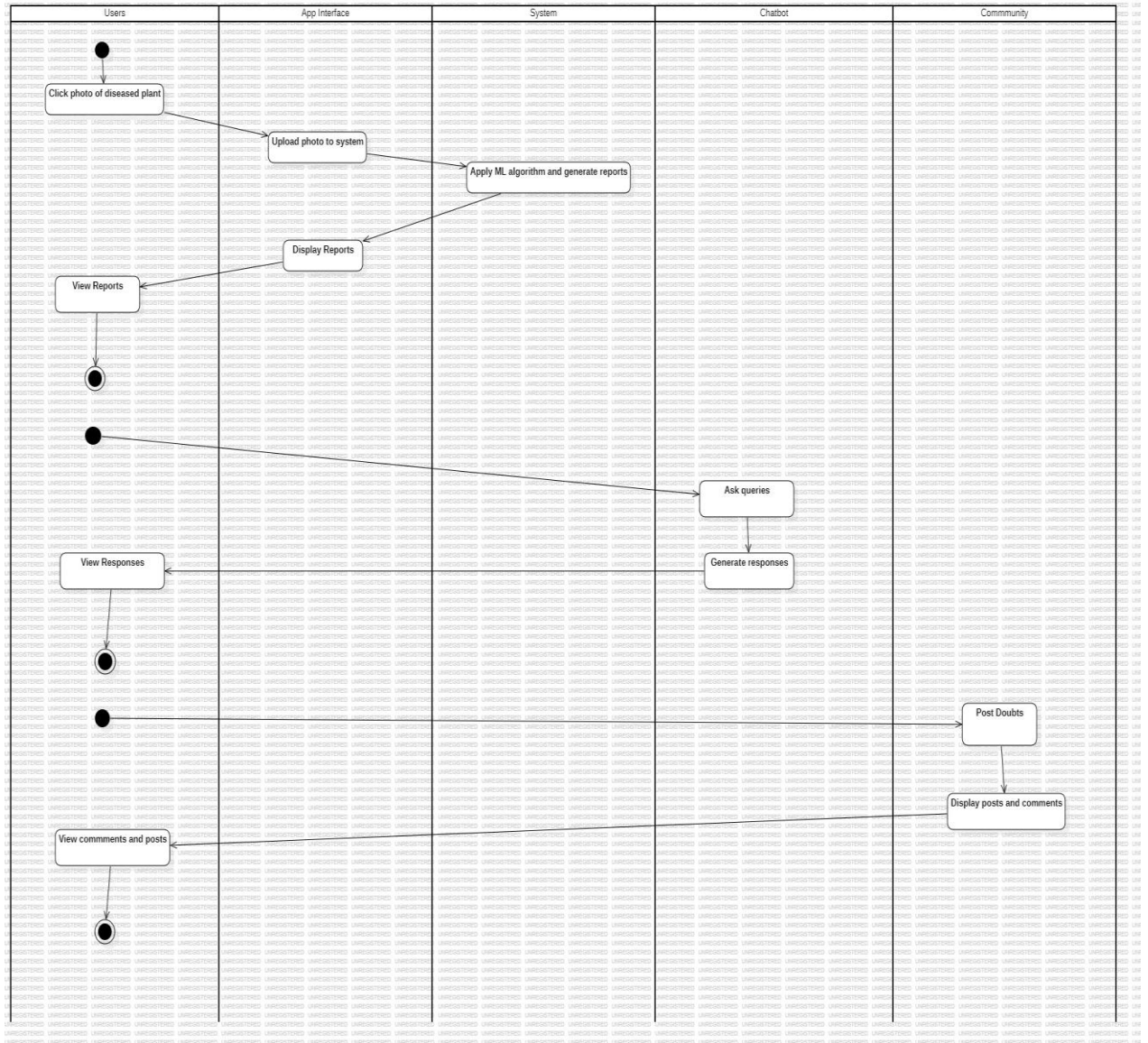


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Conclusion:

Thus we successfully developed state transition and activity diagram.



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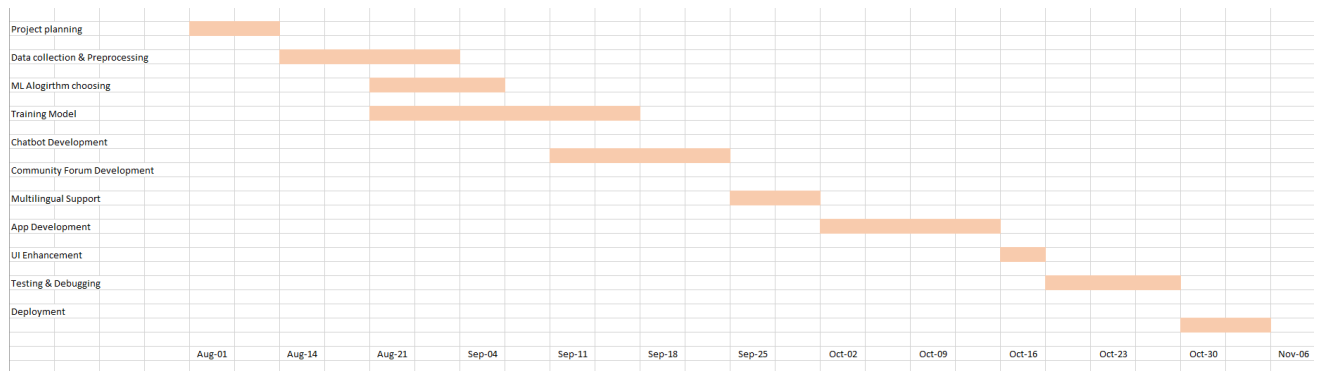
EXPERIMENT 6

Aim: Prepare Gantt Chart for IPD project.

Theory:

A Gantt chart in software engineering is a powerful visual tool used to plan and track the progress of a project over time. It displays tasks or phases of the software development process on the vertical axis and time intervals on the horizontal axis. Each task is represented by a bar whose length corresponds to the duration of the task, allowing teams to see what needs to be done, who's responsible, and when things are scheduled to start and finish. It's essential for managing complex software projects, as it helps in allocating resources, identifying dependencies between tasks, and ensuring that milestones are met on time. Gantt charts make it easier for teams to stay organized and keep track of project deadlines, making it a vital tool in Agile, Waterfall, or any other development methodology.

Gantt Chart:



Conclusion:

Thus we successfully prepared a Gantt Chart for our IPD project.



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EXPERIMENT 7

Aim: Prepare RMMM Plan.

Risk Mitigation, Monitoring and Management (RMMM) Plan

Project Title: AgriSmart – AI-powered Agriculture Assistance Platform

Prepared by: Gargi

Date: 20 April 2025

1. Objective

To ensure that the AgriSmart platform — which provides disease detection using a CNN model, multilingual support, chatbot assistance, fertilizer calculation, weather integration via APIs, and community features — remains robust, accurate, accessible, and secure throughout its development and deployment.

2. Introduction to RMMM

The RMMM plan addresses the identification, analysis, monitoring, and response to risks during the development and operational lifecycle of AgriSmart. It focuses on issues such as model inaccuracies, data dependency, integration failures, security loopholes, and user experience challenges, especially considering multilingual and rural user bases.

3. What is RMMM?

- **Risk Mitigation:** Actions to reduce the probability or impact of potential risks.
 - **Risk Monitoring:** Tracking known risks and identifying new ones throughout development.
 - **Risk Management:** Procedures to respond effectively when a risk occurs.
-

4. Why This RMMM Plan is Needed



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Since AgriSmart is AI-driven, multilingual, and dependent on APIs and third-party services, a failure in any component can directly affect usability, trust, and outcomes for farmers. A tailored RMMM plan ensures we:

- Maintain high model accuracy for disease prediction
 - Keep APIs (weather, etc.) functional and resilient
 - Protect user data (especially in community forums and chatbot)
 - Deliver a consistent experience in both English and Hindi
 - Ensure accessibility for a largely non-tech-savvy user base
-

5. Use Case: AgriSmart Overview

AgriSmart is a software platform that empowers farmers with:

- **Plant disease detection** using a trained CNN model
 - **Multilingual chatbot** support for agriculture-related queries
 - **Fertilizer recommendation calculator**
 - **Weather insights** using external APIs
 - **Community forums** for peer-to-peer interaction and knowledge sharing
 - **Support for English and Hindi**
-

6. Benefits of RMMM in AgriSmart

- Reduces chances of misinformation due to AI errors
 - Ensures accessibility across different languages
 - Guards against data misuse or cyber-attacks
 - Maintains trust and reliability through stable API integration
 - Supports scalability and long-term adoption
-

7. Challenges in Implementing RMMM

- Model accuracy may degrade over time (concept drift)
 - API limitations (rate limits, outages, outdated data)
 - Handling offensive or incorrect content in community/chatbot
 - UI/UX inconsistencies across languages
 - Limited digital literacy among end users
-

8. RMMM Risk Information Table



SHRI VILEPARLE KELAVANI MANDAL'S
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC ACCREDITED with "A" GRADE (CGPA : 3.18)



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

Risk ID	Date	Probability	Impact	Description	Mitigation / Monitoring	Management / Contingency Plan	Assigned To
AGS-01	20/04/2025	Medium (60%)	High	CNN model misclassifies plant diseases or performs poorly on unseen images	Use diverse dataset, include feedback loop, retrain periodically	Flag images with low confidence, show "Not sure" disclaimer, allow user to report wrong output	ML Lead
AGS-02	20/04/2025	High (75%)	Medium	Weather API is down or returns inaccurate data	Use fallback API provider, cache last known data	Notify users of outdated data, suggest manual verification	Backend/API Dev
AGS-03	20/04/2025	Medium (50%)	High	Users post harmful, misleading, or spammy content in community forums	Add content moderation, keyword filters, report/block system	Temporarily ban spammers, notify moderators, auto-hide flagged posts	Community Manager
AGS-04	20/04/2025	Medium (40%)	Medium	Chatbot fails to understand input in Hindi or responds incorrectly	Improve NLU model for Hindi, expand training corpus	Provide "Was this helpful?" feedback, fallback to FAQs	NLP/Chatbot Dev
AGS-05	20/04/2025	Low (30%)	High	Data breach or leakage of personal information from forums/chatbot	Use HTTPS, authentication, encrypted storage, no sensitive info stored	Notify affected users, follow data breach protocols	Security Lead
AGS-06	20/04/2025	High (80%)	Medium	Fertilizer calculator gives poor recommendations due to input error or wrong formula	Add input validation, show examples, get formulas validated by agronomist	Log edge cases, allow feedback, add expert correction pipeline	Backend/Domain Expert
AGS-07	20/04/2025	Low (20%)	Medium	English-Hindi translations create	Work with native language	Allow user to switch language on every screen easily	UI/UX Lead



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Risk ID	Date	Probability	Impact	Description	Mitigation / Monitoring	Management / Contingency Plan	Assigned To
				confusion or lose meaning	reviewers, use contextual NLP translation models		

9. Detailed Risk Breakdown Example

Risk ID: AGS-01

Description:

The CNN model may incorrectly classify plant diseases, especially if lighting, background, or crop variety differs from training data.

Mitigation / Monitoring:

- Train on diverse, high-resolution datasets (multiple seasons, varieties)
- Include a "Not sure?" option when model confidence < 50%
- Set up a retraining pipeline based on user feedback and reported errors
- Monitor model performance using F1 score and confusion matrix

Management / Contingency Plan:

- If consistent error patterns are found, hotfix the model or disable affected categories temporarily
- Notify users that detection results are indicative, not prescriptive
- Provide links to nearby agriculture support centers or contact lines

Assigned To: ML Developer (CNN Specialist)

Conclusion:

Thus we successfully prepared a RMMM plan for IPD project.



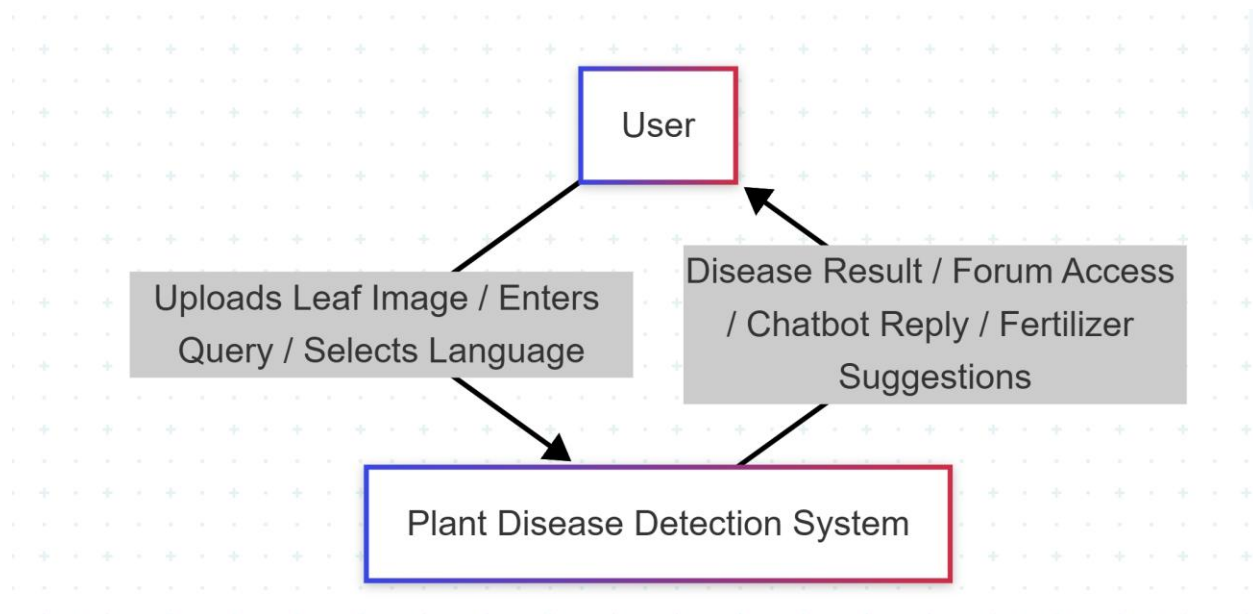
Name: Gargi Sukhatankar
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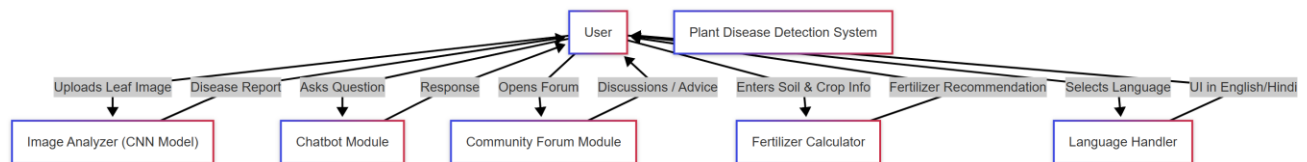
EXPERIMENT 8

Aim: To Develop and Draw Data Flow Diagram.

Level 0



Level 1



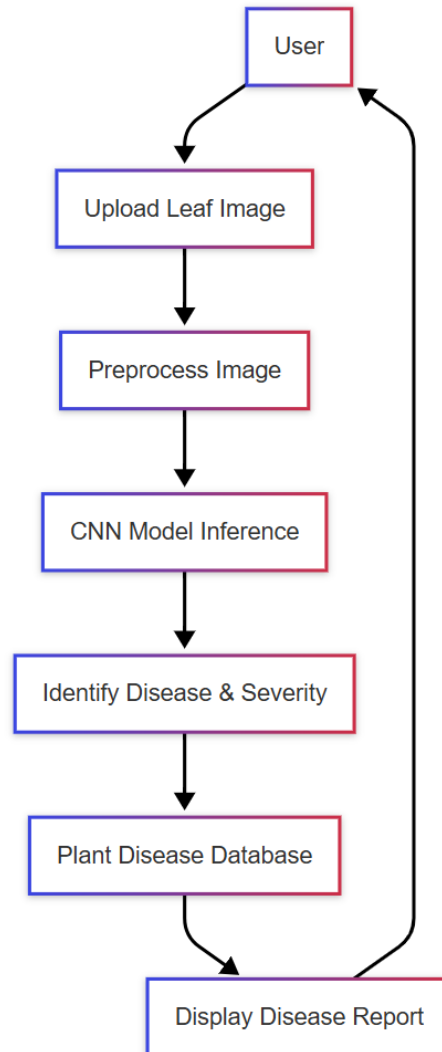
Level 2

Disease Dtection



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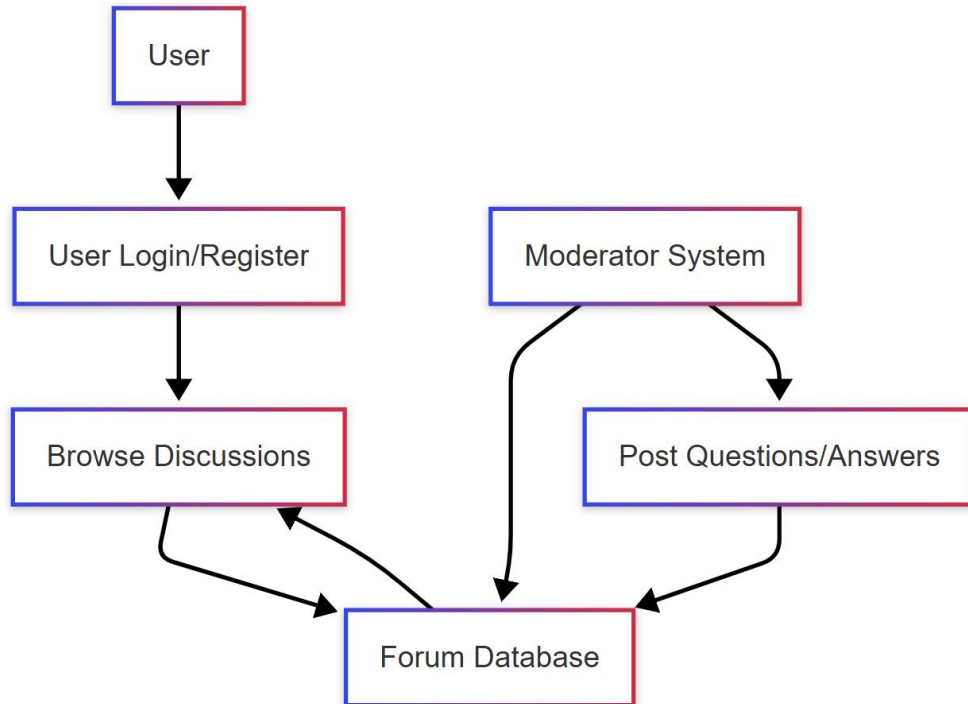


Community Forum



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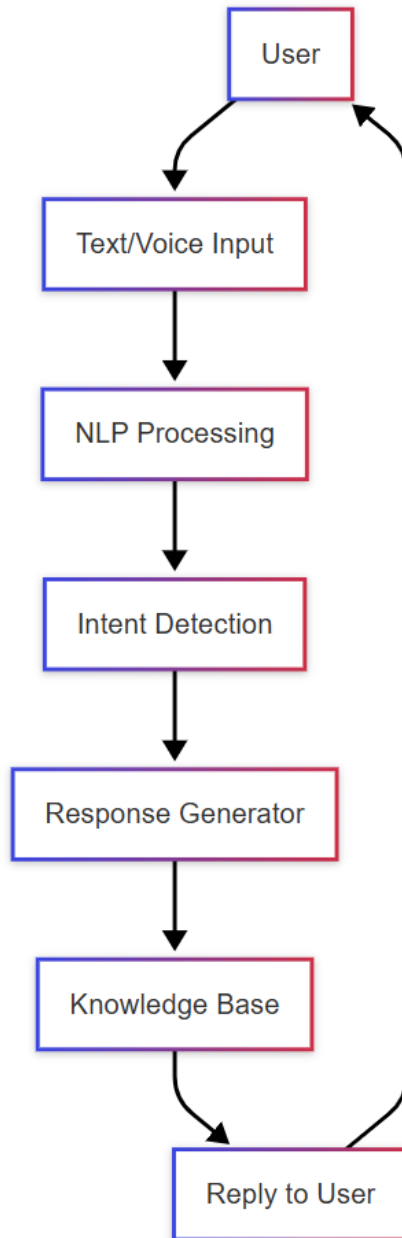


Chatbot Module



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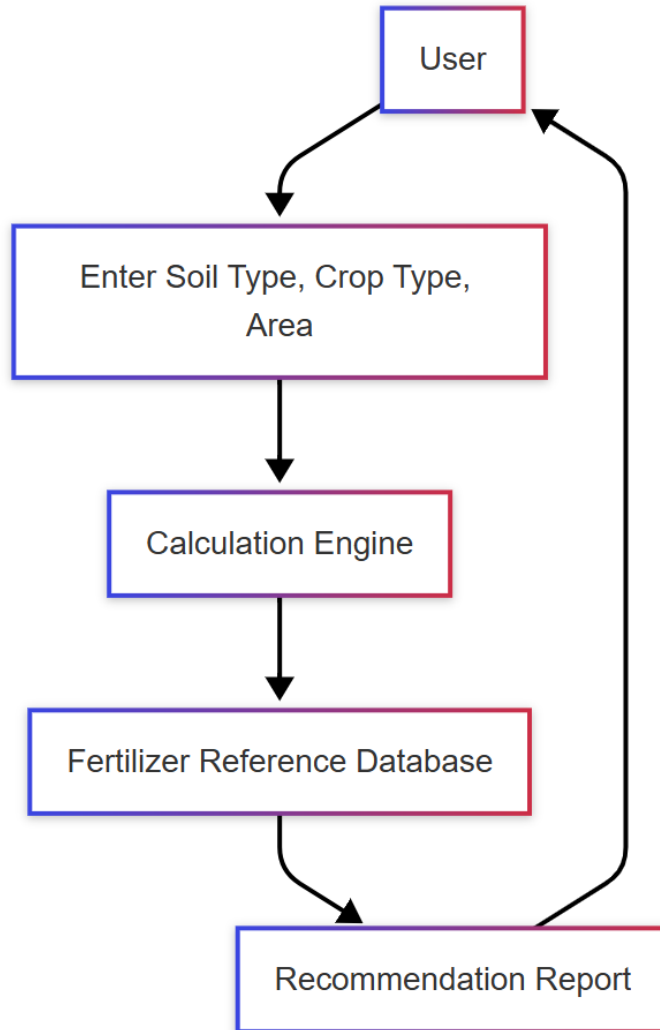


Fertilizer Calculator



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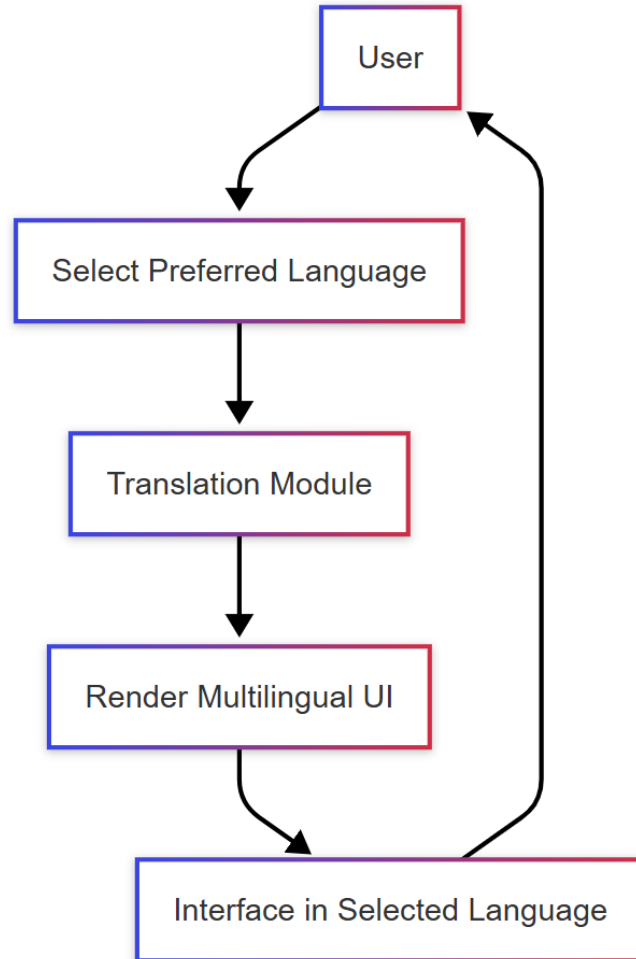


Language Support



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Conclusion: Thus we successfully developed and drew Data Flow Diagram for our IPD project.

Name: Gargi Sukhatankar

Roll No: B027

SAP ID: 60019220125

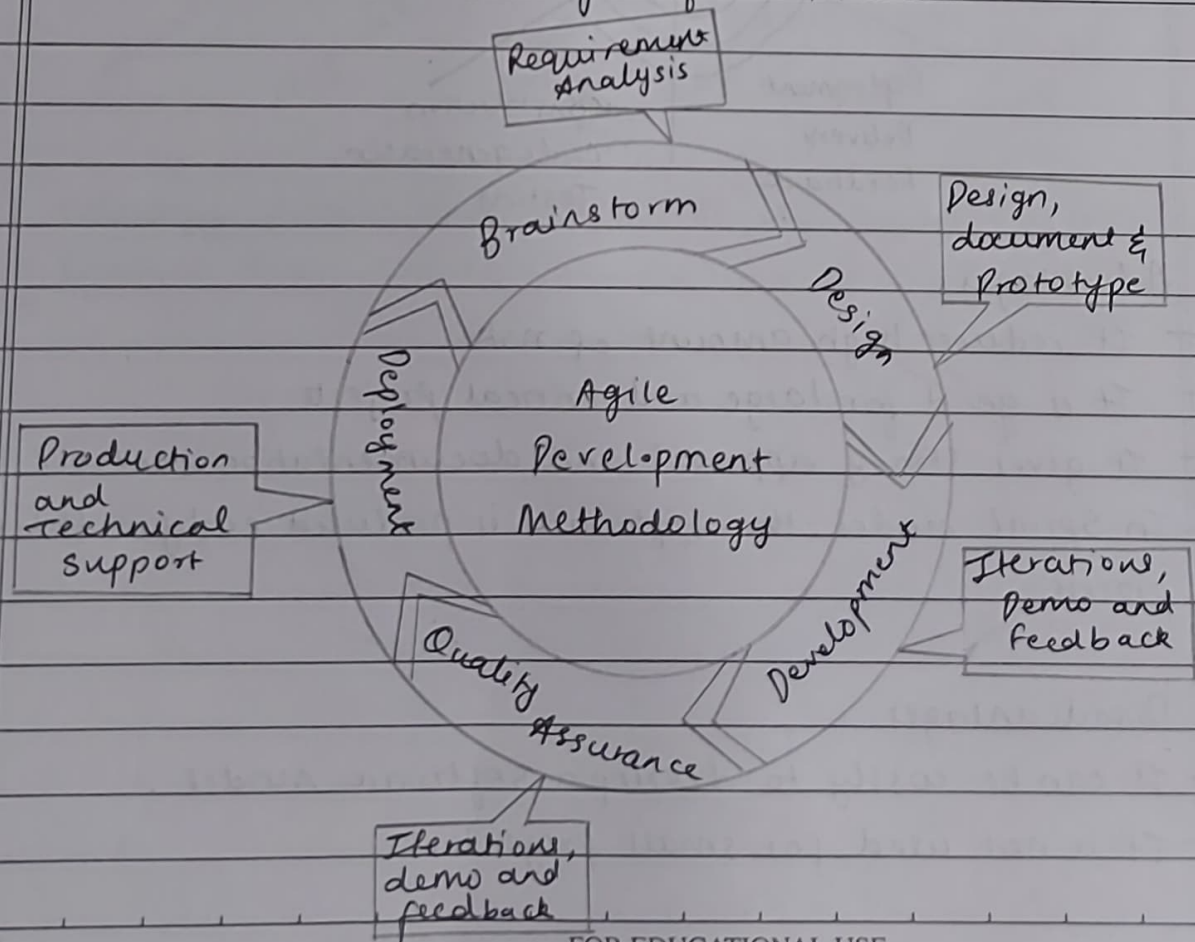
Branch: CSE (TCB)

ASSIGNMENT 1

Q-1. Suppose you are developing a project where you have to give continuous feedback to users. Which process model you will prefer?

⇒ I will prefer agile model as it is based on iterative development.

- Agile methods break tasks into smaller iterations or parts do not directly involve long term planning.
- Project scope and requirements are laid down at the beginning of the development process.
- Plans regarding the no. of iterations, the duration & scope of each iteration are clearly defined in advance.



Q-2. Explain which process model considers risk factors

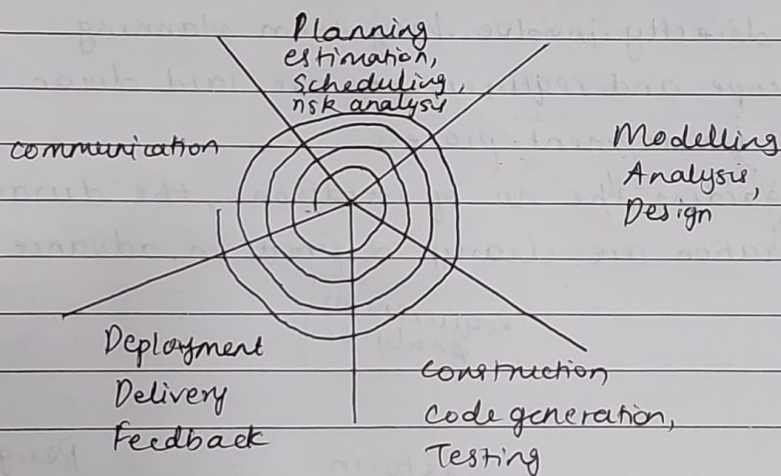
→ The Spiral Model is a risk driven process model.

→ It is used for generating the software projects.

→ In spiral model, an alternate solution is provided if the risk is found in the risk analysis, then alternate solutions are suggested & implemented.

→ It is a combination of prototype and sequential model or waterfall model.

→ In one iteration all activities are done, for large project's the output is small.



Advantages

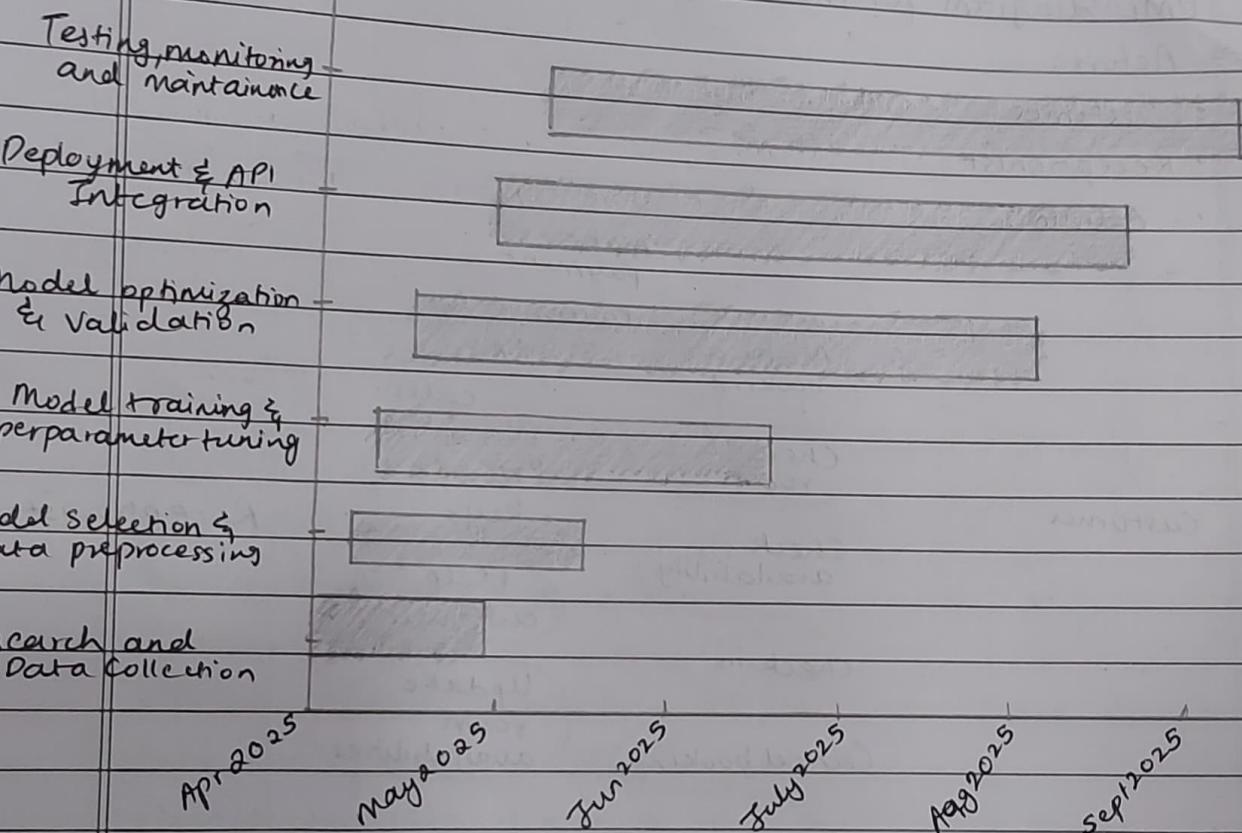
- It reduces high amount of risks
- It is good for large and critical projects
- It gives strong approval and documentation control.
- In Spiral model, the software is produced early in the life cycle process

Disadvantages

- It can be costly to develop a software model.
- It is not used for small projects.

Q-3

Prepare timeline chart to implement weather prediction model.

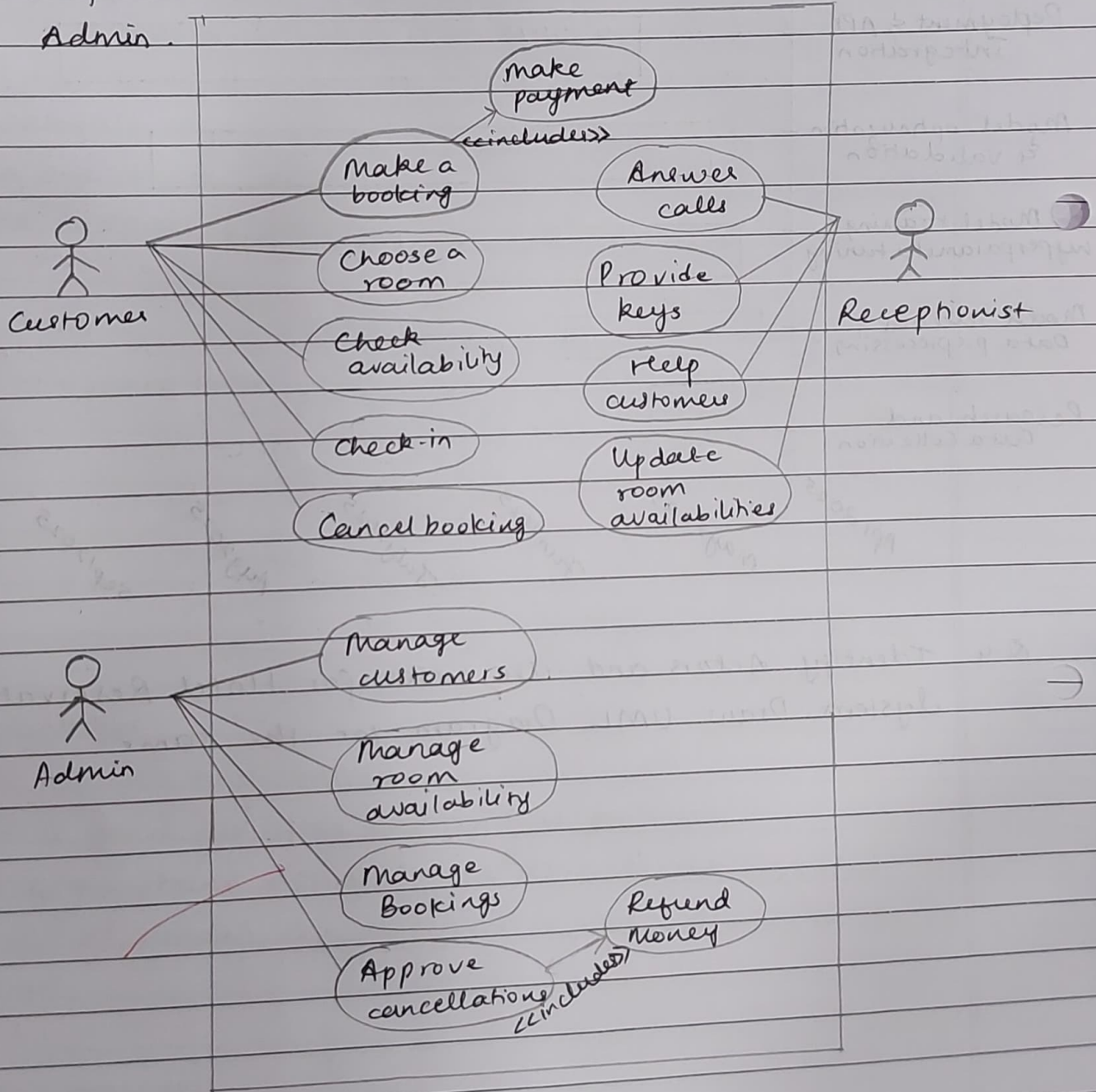


Q-4. Identify actors and use cases for Hotel Reservation System. Draw UML Diagram for the same.

Identify actors & use cases for hotel reservation system. Draw UML diagram for the same.

⇒ Actors:

- Customer
- Receptionist
- Admin.



Write a short note on walkthrough and FTR.

- ⇒ The walkthrough is a review meeting but it is different from inspection, as it does not involve any formal process i.e. it is a nonformal process.
- Basically, the walkthrough is started by the author of the code.
- In the walkthrough, the code or document is read by the author and others who are present in the meeting can be note down the important points or can write notes on the defects and can give suggestions about them.
- Walkthrough is an informal way of testing, no formal authority is been involved in this testing.
- There is no need of a moderator while performing a walkthrough.
- We can call a walkthrough an open-ended discussion, it does not focus on the documentation.
- Defect tracking is one of the challenging tasks in walkthrough.

⇒ Formal Technical Review (FTR) is a Software quality control activity performed by software engineers.

- It is an organized, methodical process for assessing and raising the standard of any technical paper, including software objects.
- Finding flaws, making sure standards are followed and improving the product or document under review's overall quality are the main objectives of a formal technical review.
- Although FTR's are frequently utilized in software development, other technical fields can also employ the concept.

Objectives of FTR:

- ① Defect Identification
- ② Quality Assurance
- ③ Risk Mitigation
- ④ Knowledge Sharing
- ⑤ Consistency and Compliance
- ⑥ Learning and Training

- In addition, FTR enables junior engineers to observe the analysis, design, coding and testing approaches closely.
- FTR also works to promote backup & continuity to become familiar with parts of the software they might not have seen otherwise.
- FTR is a class of reviews that include walkthrough, inspections, round-robin reviews and other small group technical assessments of software.
- Each FTR is conducted as a meeting and is considered successful only if it is properly planned, controlled & attended.

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SSE ASSIGNMENT 2

Q.1 Which test is performed after Unit Testing. Explain in detail.

- This type of testing is performed by developers before the setup is handed over in the testing team to formally execute the test cases.
- Unit testing is performed by the respective developers on the individual units of source code assigned areas.
- The developers use test data that is different from the test data of the quality assurance team.
- The goal of unit testing is to isolate each part of the program and show that individual parts are correct in terms of requirements and functionality.

Limitations of Unit Testing

- Testing cannot catch each and every bug in an application.
- It is impossible to evaluate every execution path in every software application. The same is the case with unit testing.
- There is a limit to the number of scenarios and test data that a developer can use to verify a source code.
- After having exhausted all the options, there is no choice but to stop unit testing and merge the code segment with other units.

Q.2 Write the test cases for login page.

⇒ Test cases are a set of conditions or input used to verify whether a system or component works as expected.
 For a login page, testing ensures secure & valid user authentication, usability and error handling.

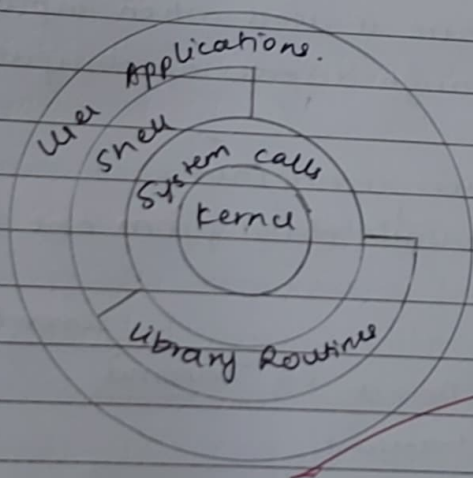
Test case ID	Test case objective	Input data	Expected result	Actual result	Status
TC1	Check cursor position at email or mobile number field.	Click on email or mobile number field.	Cursor should be placed on the field	Placed the cursor on the field	Pass
TC2	Check cursor position at password field	Click on password field	Cursor should be placed on the password field	Placed the cursor on the password field.	Pass
TC3	Check the continue button	It should redirect to password page	It redirected to the password page	It redirected to the password page	Pass
TC4	Readability of font	Try to read content on login page	Contents should be readable	Contents are readable	Pass
TC5	Testing of	check the spelling of login	Login Spelling should	Spelling of Login	Pass

Q-3.

Discuss various architecture design.

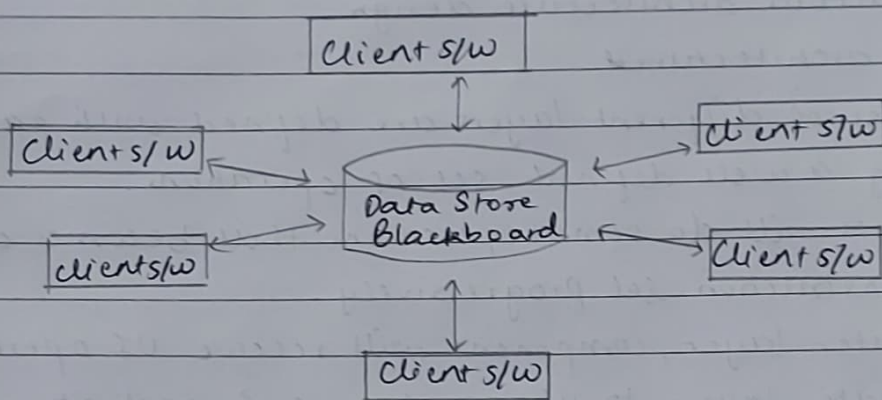
⇒ (1) Layered architecture

- A number of different layers are defined with each layer performing a well defined set of operations.
- Each layer will do some operations that become closer to machine instruction set progressively.
- At the outer layer, component will receive UI operations.
- Intermediate layers to utility services & application software functions.



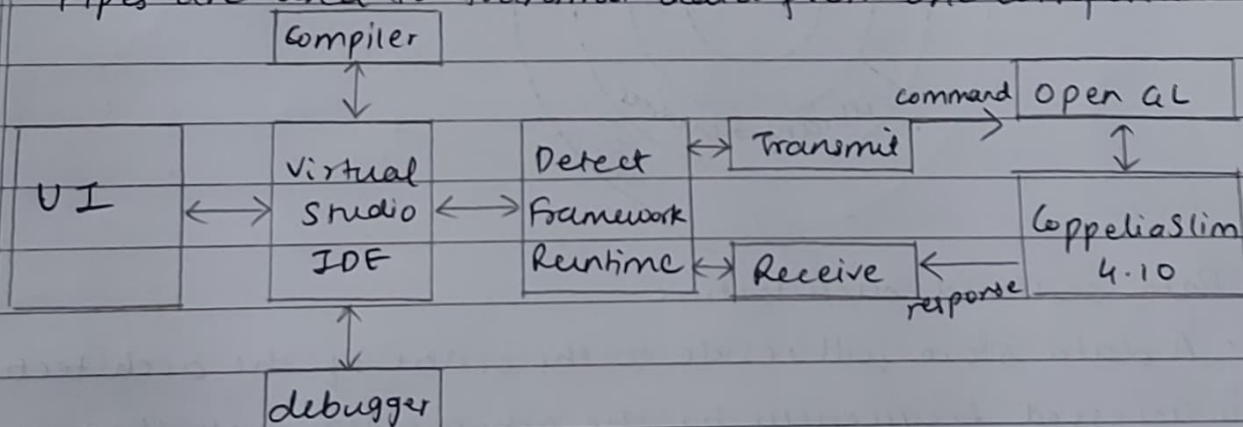
(2) Data centered architecture

- A data store will reside to the centre of the architecture & is accessed frequently by the other components that update, add, delete or modify the data.
- The client software accesses a central repository.
- Variation of this approach are used to transform the repository into a blackboard when data related to client or data of interest for the client.
- Data can be passed among clients using blackboard mechanism.



③ Data Flow architecture.

- This kind of architecture is used when input data is transformed into output data through a series of computational manipulative components.
- This is a pipe and filter architecture.
- Pipes are used to transmit data from one component to next.



Q.4. Explain the importance of cohesion and coupling in secure software engineering and their types.

- ⇒ ① Cohesion refers to how closely the operations within a single module or class are related to each other. It measures the degree to which the elements within a module work together to achieve a single, well-defined task. Following are the importance of cohesion:

- Improved readability & Maintainability: High cohesion often leads to modules that are easier to understand, modify and test.
- Reduced risk of security Flaws: When a module is tightly focused on a single responsibility, it's less likely to introduce unattended interactions.
- Easier updates and patching. A module with high cohesion can be more easily updated or patched without affecting other areas of the system.

② Coupling Types of Cohesion:

- 1) Functional Cohesion: different elements cooperate to achieve a single function.
- 2) Sequential cohesion: If element of a module form the components of the sequence, where the output from one component of the sequence is input to the next.
- 3) Communicational Cohesion: A module is said to have communicational cohesion if all tasks of the module refer to or update the same data structure.
- 4) Procedural Cohesion: If set of purpose of the module are all part of a procedure.
- 5) Temporal Cohesion: Includes ~~functions~~ that are associated by the fact that all methods must be executed.
- 6) Logical Cohesion: If elements of a module perform a similar operation.
- 7) Coincidental Cohesion: A module is said to have coincidental cohesion if it performs a set of tasks that are associated with each other very loosely.

② Coupling: Coupling refers to the degree of dependence between different modules or components in a software system.

Importance

Types of Coupling:

- Isolation of Security Risks: Ensures that failures don't cascade into others.
- Easier Security audits and testing: Easier to audit, test and validate with loose coupling.
- Simplified Maintenance: Updates and changes can be made to one module without affecting others.

Types of Coupling

- 1) No direct Coupling: Modules are subordinates to different modules. No direct coupling is present.
- 2) Data Coupling: Data is passed to another module.
- 3) Stamp Coupling: 2 modules are stamp coupled if they communicate using composite data items.
- 4) Control Coupling: Data from one module is used to direct the structure of instruction execution in another.
- 5) External Coupling: 2 modules share an externally imposed data format.
- 6) Common Coupling: They share information through some global data items.

5. Explore pointer based vulnerability with example.

- ⇒ Pointer based vulnerabilities are crucial security flaws that occur in programs written in languages like C++, where direct memory access is permitted. These vulnerabilities arise when pointer variables that store memory addresses are misused or improperly managed. Such vulnerabilities can lead to system crashes, data corruptions or even allow attackers to execute arbitrary code.

• **Buffer Overflow:** It occurs when a program writes more data into a memory buffer than it can hold. Since memory buffers are often allocated with a fixed size, any excess data can overcome adjacent memory. This may include sensitive data, or control information like return addresses on the call stack. **Risk:** An attacker could overwrite the return address of a function. Also there can be unauthorized access.

• **Null Pointer Access:** ^{Reference:} When an attempt to access or modify memory using a pointer that has not been initialized, it can cause a crash. This is a common cause of segmentation faults in programs.

Risks: While often resulting in a crash, if combined with other vulnerabilities this can be a part of a larger exploit.